

第三章. 晶格动力学和晶体的热学性质

§ 3.1 简正模和格波.

简谐近似:

$$U(x) = U(x_0) + \frac{\partial U}{\partial r} \big|_{r=r_0} (r-r_0) + \frac{1}{2} \frac{\partial^2 U}{\partial r^2} \big|_{r=r_0} (r-r_0)^2$$

一、简正模

晶体中有 N 个原子

$$H = T + V(q_1, q_2, \dots, q_N)$$

$$= \frac{1}{2} \sum_i \dot{q}_i^2 + V_0 + \sum_i \left(\frac{\partial V}{\partial q_i} \right)_0 q_i + \frac{1}{2} \sum_i \sum_j \frac{\partial^2 V}{\partial q_i \partial q_j} q_i q_j$$

$$= \frac{1}{2} \sum_i \dot{q}_i^2 + \frac{1}{2} \sum_{ij} \lambda_{ij} q_i q_j$$

引入位移矢量:

$$H = \frac{1}{2} \dot{q}^T \dot{q} + \frac{1}{2} q^T \lambda q$$

$$\exists A, A^T \lambda A = \text{diag}(\omega_i^2) \text{ 且 } A^T A = 1$$

$$H = \frac{1}{2} \dot{q}^T A A^T \dot{q} + \frac{1}{2} q^T A \text{diag}(\omega_i^2) A^T q$$

$$= \frac{1}{2} \dot{Q}^T \dot{Q} + \frac{1}{2} Q^T \text{diag}(\omega_i^2) Q$$

$$Q = A^T q \quad / \quad q = A Q$$

$$\text{Euler 方程: } \ddot{Q} + \text{diag}(\omega_i^2) Q = 0$$

$$\Rightarrow Q_j = C e^{-i\omega_j t}$$

$$q_i = \text{Re} \left[\sum_j a_{ij} Q_j \right] = \text{Re} \left[\sum_j a_{ij} C e^{-i\omega_j t} \right]$$

在量子理论中:

$$P_i = -i\hbar \frac{\partial}{\partial Q_i}$$

$$\text{系统的 Schrodinger 方程为 } \frac{1}{2} \sum_j (-\hbar^2 \frac{\partial^2}{\partial Q_j^2} + \omega_j^2 Q_j^2) \psi(Q) = E \psi(Q)$$

↓ 分离变量

$$\frac{1}{2} (-\hbar^2 \frac{\partial^2}{\partial Q_j^2} + \omega_j^2 Q_j^2) \psi(Q_j) = E_j \psi(Q_j)$$

解为 $\varepsilon_j = (n_j + \frac{1}{2}) \hbar \omega_j$

$$\varphi_{n_j}(\alpha_j) = \frac{1}{\sqrt{\hbar 2^n n_j!}} \left(\frac{\omega_j}{\hbar} \right)^{n_j/2} \exp \left(-\frac{\omega_j}{2} \right) H_{n_j} \left(\sqrt{\frac{\omega_j}{\hbar}} \alpha_j \right)$$

其中 $\alpha_j = \sqrt{\frac{\omega_j}{\hbar}} \alpha_j$

∴ $E = \sum_j \varepsilon_j = \sum_j (n_j + \frac{1}{2}) \hbar \omega_j$

$\psi = \prod_j \varphi_{n_j}(\alpha_j)$

二、格波

一个简正模 $u_s(\vec{r}) = u(\vec{r}) e^{-i\omega_j t}$, 考虑晶体平移对称性对其施加的限制.

$$\vec{r} = l_1 \vec{a}_1 + l_2 \vec{a}_2 + l_3 \vec{a}_3$$

$u(\vec{r}) = A_{j\vec{r}}$ $\vec{r} = 1, 2, 3$ 为三个振动的方向.

由于平移对称性, 其它位置振幅只与 $A_{j\vec{r}}$ 差相位 $\lambda(\vec{r})$

$$u(\vec{a}_1) = \lambda(\vec{a}_1) u(\vec{r}), \quad u(l_1 \vec{a}_1) = \lambda(\vec{a}_1)^{l_1} u(\vec{r})$$

$$u(\vec{a}_2) = \lambda(\vec{a}_2) u(\vec{r}), \quad u(l_2 \vec{a}_2) = \lambda(\vec{a}_2)^{l_2} u(\vec{r})$$

$$u(\vec{a}_3) = \lambda(\vec{a}_3) u(\vec{r}), \quad u(l_3 \vec{a}_3) = \lambda(\vec{a}_3)^{l_3} u(\vec{r})$$

∴ $u(\vec{r}) = \lambda(\vec{a}_1)^{l_1} \lambda(\vec{a}_2)^{l_2} \lambda(\vec{a}_3)^{l_3} u(\vec{r})$

⇒ $|\lambda(\vec{a}_1)| = |\lambda(\vec{a}_2)| = |\lambda(\vec{a}_3)| = 1$

$$\lambda(\vec{a}_i + \vec{a}_j) = \lambda(\vec{a}_i) \lambda(\vec{a}_j)$$

因此:

$$\lambda(\vec{a}_i) = e^{i\vec{q} \cdot \vec{a}_i} \quad i=1, 2, 3.$$

$$u(\vec{r}) = e^{i\vec{q} \cdot \vec{r}} u(\vec{r}) = A_{j\vec{r}} e^{i\vec{q} \cdot \vec{r}}$$

在晶格中的波动解:

$$A_{j\vec{r}} e^{i\vec{q} \cdot \vec{r} - i\omega_j t} = A_{j\vec{r}} e^{i(\vec{q} \cdot \vec{r} - \omega_j t)}$$

一个包含 $3N$ 个自由度的周期性结构, 存在 $3N$ 个独立的简正模

等价于 $3N$ 个独立的格波

§ 3.1 - 单原子链的振动

一、运动方程

质量为 m (平衡) 间距 a , 位移 u_n

$$m \frac{d^2 u_n}{dt^2} = \sum_p \beta_p (u_{p+n} - u_n)$$

只考虑最近邻

$$m \frac{d^2 u_n}{dt^2} = \beta_n (u_{n+1} - u_n) + \beta_{n-1} (u_n - u_{n-1})$$

平移对称性: $\beta_{n+1} = \beta_{n-1} = \beta$

$$m \frac{d^2 u_n}{dt^2} = \beta (u_{n+1} + u_{n-1} - 2u_n)$$

代入格波解: $u_n = A e^{i(qna - \omega t)}$

$$-m\omega^2 A = \beta A (e^{iqa} + e^{-iqa} - 2)$$

$$\hookrightarrow \text{色散关系: } \omega = 2\sqrt{\frac{\beta}{m}} |\sin(\frac{1}{2}qa)|$$

连续介质: 取长波极限. $\lambda \gg a \rightarrow q = \frac{2\pi}{\lambda} \ll \frac{2\pi}{a} \ll \frac{2\pi}{a}$

$$\omega = \sqrt{\frac{\beta}{m}} a q = cq.$$

二、格波特性

群速度: $v_g = \frac{\partial \omega}{\partial q} = \sqrt{\frac{\beta}{m}} a \cos(\frac{1}{2}qa)$

相速度: $v_p = \frac{\omega}{q} = 2\sqrt{\frac{\beta}{m}} \frac{|\sin(\frac{1}{2}qa)|}{a}$

周期性: $\omega(q) = \omega(q + \frac{2\pi}{a}n) = \omega(q + \frac{2\pi}{a})$

应写倒格矢.

因此所有独立的振动模式都在第一布里渊区内.

$$q \in (-\frac{\pi}{a}, \frac{\pi}{a}]$$

在 $q = \frac{\pi}{a}$ 的边界: $u_n = A e^{i\pi n - i\omega t} = (-1)^n A e^{-i\omega t}$

$$v_g = \sqrt{\frac{\beta}{m}} a \cos(\frac{1}{2} \frac{\pi}{a} a) = 0$$

表现为驻波.

$\omega \in [0, \omega_{max}]$ 低通滤波器.

三. Born-Karman 边界条件.

$$1. A e^{i(qLa - wt)} = A e^{i(q(1+N)a - wt)}$$

$$\hookrightarrow e^{iqNa} = 1, \quad qNa = 2\pi h \rightarrow q = \frac{2\pi h}{Na}, \quad h \in \mathbb{Z}$$

在第一布里渊区 $-\frac{\pi}{a} < q \leq \frac{\pi}{a}$ 内, $-\frac{N}{2} < h \leq \frac{N}{2}$

有 N 个独立模式 / 波矢 (N 个简谐波, 均匀分布)

$$\text{波矢密度: } \frac{N}{2\pi} = \frac{Na}{2\pi}$$

2. 所有独立振动模式正交完备.

$$\text{正交性: } \frac{1}{N} \sum_{l=0}^{N-1} e^{i(q-q')la} = \delta_{qq'}$$

$$\text{证: 令 } \phi = (q - q')a$$

$$\text{原式} = \frac{1}{N} \frac{1 - e^{iN\phi}}{1 - e^{i\phi}} = \frac{1}{N} \frac{1 - e^{iqNa - iq'Na}}{1 - e^{i\phi}} \quad \uparrow = 0$$

$$= \begin{cases} 0 & (q \neq q') \\ \frac{1}{N} \cdot N = 1 & (q = q') \end{cases} = \delta_{q,q'}$$

$$\text{完备性: } \frac{1}{N} \sum_q e^{iq(1-l')a} = \delta_{ll'}$$

$$\sum_k (V_k < V_k) = I$$

$$\text{证: } \frac{1}{N} \sum_q e^{i \frac{2\pi h}{Na} \cdot (1-l')a}$$

$$= \frac{1}{N} \sum_{h=-\frac{N}{2}}^{\frac{N}{2}} e^{2\pi i \frac{h}{N} (1-l')} \quad , \quad \text{令 } \phi = \frac{2\pi(1-l')}{N}$$

$$= \frac{1}{N} \frac{e^{i h_{\min} \phi} (1 - e^{iN\phi})}{1 - e^{i\phi}} = \begin{cases} 0 & (1 \neq l') \\ \frac{1}{N} \cdot N = 1 & (1 = l') \end{cases}$$

$$e^{i h_{\min} \phi} = 1 \quad (1 = l')$$

四. 简正坐标

1. 简征解: $u_{iq} = A_{iq} e^{i[q/a - \omega(q)t]}$

$$u_i = \sum_q u_{iq} = \sum_q A_{iq} e^{i[q/a - \omega(q)t]}$$

定义约化坐标 $q_2 = \sqrt{m} u_i$, $Q_q(t) = \sqrt{m} A_{iq} e^{-i\omega(q)t}$

$$q_1 = \sum_q \frac{1}{\sqrt{N}} e^{iq_1 a} \cdot Q_q(t) \quad \vec{q} = \mathbf{A} \cdot \mathbf{Q}$$

$$\mathbf{A} = \langle A_{iq} \rangle = \sum_q \frac{1}{\sqrt{N}} e^{iq_1 a}$$

2. $H = T + V = \frac{1}{2} \sum_i \left[m \dot{u}_i^2 + \beta (u_{i+1} - u_i)^2 \right]$

$$u_i = \frac{1}{\sqrt{Nm}} \sum_q e^{iq_1 a} Q_q(t)$$

u_i 为实坐标: $u_i = u_i^* \rightarrow Q_q^* = Q_{-q}$

$T = \frac{1}{2} m \sum_i \dot{u}_i^2 = \frac{1}{2N} \sum_i \sum_q \sum_{q'} \dot{Q}_q \dot{Q}_{q'} e^{i(q+q')a}$ 正交性.

$$\frac{1}{N} \sum_q e^{i(q-q')a} = \delta_{qq'}$$

$$= \frac{1}{2} \sum_{q, q'} \dot{Q}_q \dot{Q}_{q'} \delta_{q, q'} = \frac{1}{2} \sum_q \dot{Q}_q \dot{Q}_{-q} = \frac{1}{2} \sum_q \dot{Q}_q \dot{Q}_q^*$$

$$V = \frac{1}{2} \beta \sum_i (u_{i+1} - u_i)^2$$

$$= \frac{1}{2} \beta \cdot \sum_i \frac{1}{Nm} \left[\sum_q e^{iq_1 a} (e^{iq_2 a} - 1) Q_q(t) \right]^2$$

$$= \frac{\beta}{2Nm} \sum_i \sum_q e^{iq_1 a} (e^{iq_2 a} - 1) Q_q(t) \sum_{q'} e^{iq'_1 a} (e^{iq'_2 a} - 1) Q_{q'}(t)$$

$$= \frac{\beta}{2Nm} \sum_{q, q'} e^{i(q+q')a} (e^{i(q+q')a} - e^{iq_2 a} - e^{iq'_2 a} + 1) Q_q(t) Q_{q'}(t)$$

$$= \frac{\beta}{2m} \sum_{q, q'} (e^{i(q+q')a} - e^{iq_2 a} - e^{iq'_2 a} + 1) Q_q(t) Q_{q'}(t) \delta_{q, q'}$$

$$= \frac{\beta}{2m} \sum_q (2 - e^{iq_2 a} - e^{-iq_2 a}) Q_q(t) Q_{-q}(t)$$

$$= \frac{\beta}{m} \sum_q [1 - \cos(q_2 a)] Q_q(t) Q_q^*(t)$$

$$= \frac{1}{2} \sum_q \omega^2(q) Q_q Q_q^*, \text{ 其中 } \omega(q) = 2 \sqrt{\frac{\beta}{m}} \sin^2(\frac{1}{2} q_2 a)$$

$$H = \frac{1}{2} \sum_q \alpha_q \alpha_q^\dagger + \omega^2 \alpha_q \alpha_q^\dagger, \text{ 在简正坐标下对角化}$$

§ 3.3 一维双原子链振动

N个元胞，各有M, m 两种原子，设第l个元胞中，其位移分别为 u_l, v_l

一、运动方程

$$\begin{cases} M \frac{d^2 u_l}{dt^2} = \beta (v_{l-1} + v_l - 2u_l) & \text{⊗} & \text{○} & \text{⊗} \\ & v_{l-1} & u_l & v_l \\ m \frac{d^2 v_l}{dt^2} = \beta (u_{l-1} + u_l - 2v_l) \end{cases}$$

代入试探解:
$$\begin{cases} u_l = A e^{i(qla - \omega t)} \\ v_l = B e^{i(qla - \omega t)} \end{cases}$$

$$\begin{cases} -\omega^2 M A = \beta [B (1 + e^{-iqa}) - 2A] \\ -\omega^2 m B = \beta [A (e^{iqa} + 1) - 2B] \end{cases}$$

方程有解条件:

$$\begin{aligned} m\omega^2 - 2\beta & \quad \beta (1 + e^{-iqa}) \\ \beta (e^{iqa} + 1) & \quad m\omega^2 - 2\beta \end{aligned} = 0$$

$$(m\omega^2 - 2\beta)(m\omega^2 - 2\beta) - 2\beta^2 [1 + \cos(qa)] = 0$$

$$Mm\omega^4 - 2\beta(M+m)\omega^2 + 4\beta^2 \left\{ 1 - \frac{1}{2} [1 + \cos(qa)] \right\} = 0$$

$$Mm\omega^4 - 2\beta(M+m)\omega^2 + 4\beta^2 \sin^2 \left(\frac{1}{2} qa \right) = 0$$

$$\begin{aligned} \omega_{\pm}^2 &= \frac{1}{2Mm} \left\{ 2\beta(M+m) \pm \sqrt{[2\beta(M+m)]^2 - 4Mm \cdot 4\beta^2 \sin^2 \left(\frac{1}{2} qa \right)} \right\}^{1/2} \\ &= \frac{\beta(M+m)}{Mm} \left\{ 1 \pm \sqrt{1 - \frac{4Mm \sin^2 \left(\frac{1}{2} qa \right)}{(M+m)^2}} \right\}^{1/2} \end{aligned}$$

求重原子振幅和相位比

$$\alpha_{\pm} = \left(\frac{B}{A} \right)_{\pm} = \frac{2\beta - M\omega_{\pm}^2}{\beta (1 + e^{-iqa})}$$

$$= \frac{1}{1 + e^{-iqa}} \left\{ 2 - \frac{M+m}{m} \left\{ 1 \pm \sqrt{1 - \frac{4Mm \sin^2 \left(\frac{1}{2} qa \right)}{(M+m)^2}} \right\}^{1/2} \right\}$$

二、声学波和光学波.

$$\omega_{\pm}(q) \approx \omega_{\pm}(q + \frac{2\pi h}{a}) = \omega_{\pm}(q + k)$$

1. ω_{-} 支 (声学支)

$$\begin{cases} \omega_{-min} = 0, & q = 0 \\ \omega_{-max} = \sqrt{\frac{2\beta}{m}}, & q = \frac{\pi}{a}. \end{cases}$$

$q \rightarrow 0$ 时,

$$\omega_{-}(q) = \sqrt{\frac{2\beta}{m+m}} qa = cq$$

$$\alpha_{-}(q) = \left(\frac{\beta}{A}\right)_{-} = \frac{2\beta - m \cdot \frac{2\beta}{m+m} q^2 a^2}{\beta(1 + e^{-iqa})} \approx 1 \quad \text{轻、重原子的振幅相位相同.}$$

此时 $\frac{d\omega_{-}(q)}{dq} = \frac{\omega}{q} = c$, 表现为弹性波.



$q \rightarrow \frac{\pi}{a}$ 时,

$$\omega_{-} \approx \sqrt{\frac{2\beta}{m}}$$

$$\alpha_{-} \approx \frac{2\beta - m \cdot \frac{2\beta}{m}}{\beta(1 + e^{-iqa})} = 0 \quad \text{重原子的振动, 轻原子的几乎不动.}$$



2. ω_{+} 支, 光学支.

$$\begin{cases} \omega_{+min} = \sqrt{\frac{2\beta}{m}} \\ \omega_{+max} = \sqrt{\frac{2\beta}{M}} \end{cases} \quad \begin{array}{l} \text{轻、重原子(正、负离子)相对运动引起极化} \\ \text{与远红外电磁场耦合.} \end{array}$$

$q \rightarrow 0$ 时,

$$\omega_{+} \approx \left\{ \beta \frac{(m+M)}{mM} \right\}^{1/2} \left[2 - \frac{1}{2} \frac{mM q^2 a^2}{(m+M)^2} \right]^{1/2} \approx \sqrt{\frac{2\beta}{m}}$$

$$\alpha_{+} = \left(\frac{\beta}{A}\right)_{+} = \frac{2\beta - M\omega^2}{2\beta} = 1 - \frac{m}{M} = -\frac{m}{M} \quad \text{轻、重原子反向振动}$$

$q \rightarrow \frac{\pi}{a}$ 时,

$$\omega_{+} \approx \sqrt{\frac{2\beta}{M}}$$

$$\alpha_{+} \approx \frac{2 - m \cdot \frac{m}{M}}{1 + e^{-iqa}} \approx \frac{2 - \frac{m}{M}}{2\epsilon_0} = \pm \infty \quad \text{重原子的不动, 轻原子的振动.}$$



带通滤波器: $\omega \in [0, \sqrt{\frac{2\beta}{m}}] \cup [\sqrt{\frac{2\beta}{M}}, \sqrt{\frac{2\beta}{m}}]$



三、玻恩-福卡门边界条件.

$$u_i = u_{i-N}, v_i = v_{i-N}, e^{iqn} = 1, q = \frac{2\pi h}{N a}, n \in \{-\frac{N}{2}, \frac{N}{2}\}$$

独立波矢数 = 元胞数 (N), 独立模式数 = 自由度数 ($2N$)

§ 3.4 三维晶格振动 声子

一、三维晶格振动

1. N 个元胞, 每个元胞有 n 个原子, 第 i 个元胞 $\vec{R}_i = l_1 \vec{a}_1 + l_2 \vec{a}_2 + l_3 \vec{a}_3$

$$\text{系统动能 } T = \sum_{j \in \{1, N\}} \frac{1}{2} m_j \dot{u}_{j0}^2 \quad i \in \{1, N\}, j \in \{1, n\}$$

$$\text{势能 } V = \frac{1}{2} \sum_{i, j, i', j'} \frac{\partial^2 V}{\partial u_{j0} \partial u_{j'0}} u_{j0} u_{j'0}$$

$$\text{运动方程, } m_j \ddot{u}_{j0} = - \sum_{j'0'} \lambda_{j0, j'0'} u_{j'0'}$$

$$\text{其中 } \lambda_{j0, j'0'} = \left[\frac{\partial^2 V}{\partial u_{j0} \partial u_{j'0}} \right]$$

平移对称性给出:

$$\lambda_{j0, j'0'} = \lambda_{j'0, j0} (\vec{R}_i - \vec{R}_{i'})$$

$$\Rightarrow m_j \ddot{u}_{j0} = \sum_{j'0'} \lambda_{j'0, j0} (\vec{R}_i - \vec{R}_{i'}) u_{j'0'}$$

$$\text{取平面波解 } u_{j0} = A_{j0\vec{q}} e^{i(\vec{q} \cdot \vec{R}_i - \omega t)}$$

$$m_j \omega^2 A_{j0\vec{q}} e^{i(\vec{q} \cdot \vec{R}_i - \omega t)} = \sum_{j'0'} \lambda_{j'0, j0} (\vec{R}_i - \vec{R}_{i'}) A_{j'0\vec{q}} e^{i(\vec{q} \cdot \vec{R}_{i'} - \omega t)}$$

$$m_j \omega^2 A_{j0\vec{q}} = \sum_{j'0'} \lambda_{j'0, j0} (\vec{R}_i - \vec{R}_{i'}) A_{j'0\vec{q}} e^{-i\vec{q} \cdot (\vec{R}_i - \vec{R}_{i'})}$$

$$= \sum_{j'0'} A_{j'0\vec{q}} \sum_i \lambda_{j'0, j0} (\vec{R}_i - \vec{R}_{i'}) e^{-i\vec{q} \cdot (\vec{R}_i - \vec{R}_{i'})}$$

$$= \sum_{j'0'} \lambda_{j'0, j0}(\vec{q}) A_{j'0\vec{q}} \quad \lambda_{j'0, j0}(\vec{q}) = \sum_i \lambda_{j'0, j0} \vec{R}_i e^{-i\vec{q} \cdot \vec{R}_i}$$

动量守恒。

$$\Rightarrow \sum_{j'0'} (\lambda_{j'0, j0}(\vec{q}) - m_j \omega^2 \delta_{jj'}) A_{j'0\vec{q}} = 0$$

存在非零解的条件为: $|\lambda_{j'0, j0}(\vec{q}) - m_j \omega^2 \delta_{jj'}|_{j'0} = 0$

解为 $\omega_s(\vec{q})$, $s = 1, \dots, 3n$.

3 支声学波 3 个方向上原子整体平移 / 1 支纵波

$3n-3$ 支光学波, $\begin{cases} n-1 \text{ 支纵波} \\ 2(n-1) \text{ 支横波} \end{cases}$

动力学方程具有周期性: $\lambda_{j\vec{q}}(\vec{q} + \vec{k}_n) = \lambda_{j\vec{q}}(\vec{q})$

$\downarrow$
 解有周期性: $\omega_s(\vec{q} + \vec{k}_n) = \omega_s(\vec{q})$

2. 玻恩-卡门边界条件.

设晶体在 i 个方向各有 N_i ($i=1, 2, 3$) 个元胞.

$$u_{l+N_i, j} = u_{l, j}$$

$$A_{j\vec{q}} e^{i(\vec{q} \cdot \vec{R}_{l+N_i} - \omega t)} = A_{j\vec{q}} e^{i(\vec{q} \cdot \vec{R}_l - \omega t)}$$

其中 $\vec{R}_{l+N_i} = (l_i + N_i) \vec{a}_i$

$$\Rightarrow e^{i\vec{q} \cdot N_i \vec{a}_i} = 1 \quad i=1, 2, 3, \quad \vec{q} \cdot N_i \vec{a}_i = 2\pi h_i$$

$$\hookrightarrow \vec{q} = \sum_i \frac{h_i}{N_i} \vec{a}_i, \quad h_i \in (-\frac{N_i}{2}, \frac{N_i}{2}]$$

独立波矢数 = 元胞数 ($N_1 N_2 N_3$)

$$\text{波矢密度} = \frac{N}{\Omega^*} = \frac{N \cdot \Omega}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

独立格波数 = 总自由度 ($3nN$)

二、格波量 $\rightarrow$ 声子.

格波 $\vec{q}, \omega_s(\vec{q}) \sim$ 简正坐标 $Q_{\vec{q}s}$ 谐振子.

$$\varepsilon_{\vec{q}s} = (n_{\vec{q}s} + \frac{1}{2}) \hbar \omega_s(\vec{q}), \quad n_{\vec{q}s} = 0, 1, \dots, \infty$$

在温度 T 下, $\mathcal{P}(n_{\vec{q}s} = n) = \frac{e^{-\beta n \hbar \omega_s}}{\sum_{n_{\vec{q}s}} e^{-\beta n_{\vec{q}s} \hbar \omega_s}}$
 一个振子的平均占据数

$$\langle n_{\vec{q}s} \rangle = \sum_n n \mathcal{P}(n_{\vec{q}s} = n)$$

$$= \left(- \frac{1}{\hbar \omega_s(\vec{q})} \right) \frac{\partial}{\partial \beta} \ln \left(\sum_{n_{\vec{q}s}} e^{-\beta n_{\vec{q}s} \hbar \omega_s(\vec{q})} \right)$$

$$= - \frac{1}{\hbar \omega_s(\vec{q})} \frac{\partial}{\partial \beta} \ln \left(1 - e^{-\beta \hbar \omega_s(\vec{q})} \right)$$

$$= \frac{1}{\hbar \omega_s(\vec{q})} \frac{-e^{-\beta \hbar \omega_s(\vec{q})}}{1 - e^{-\beta \hbar \omega_s(\vec{q})}} \left[-\hbar \omega_s(\vec{q}) \right]$$

$$= e^{\beta \hbar \omega_s(\vec{q})} - 1 \quad \text{服从 } \mu=0 \text{ 的玻色分布.}$$

2. 声子的准动量.

$$\begin{aligned} \text{声子的动量 } p &= m \frac{d}{dt} \sum_l u_l(t) \\ &= m \frac{d}{dt} \sum_l A e^{i(q l a - \omega t)} \\ &= m (-i\omega) A e^{-i\omega t} \sum_{l=1}^N e^{i q l a} \\ &= -i m \omega A e^{-i\omega t} \frac{e^{i q a}}{1 - e^{i q a}} (1 - e^{i q N a}) \\ &= -i m \omega A e^{-i\omega t} N \delta_{q,0} \end{aligned}$$

只有 $q=0$, $\omega(q)=0$ 对总动量有贡献

非弹性的光子散射.

$$\begin{cases} \vec{k}' - \vec{k} = \pm \vec{q} + \vec{k}_h \\ E(\vec{k}') - E(\vec{k}) = \pm \hbar \omega_s(\vec{q}) \end{cases}$$

§ 3.5 离子晶体中的长光学波.

→ 正、负离子的反向运动.

一、晶格振动产生的内场

设正、负离子晶体位移为 $\vec{u}_+$, $\vec{u}_-$

$$\vec{P} = \frac{1}{V} q^+ (\vec{u}_+ - \vec{u}_-)$$

∴

产生极化波: $\vec{P}(\vec{q}, \vec{r}, t) = \vec{P}_0 e^{i(\vec{q} \cdot \vec{r} - \omega t)}$

相应电场: $\vec{E} = \frac{\omega^2}{\epsilon_0} \vec{P} - \vec{q}(\vec{q} \cdot \vec{P})$
 $\epsilon_0(q^2 - \frac{\omega^2}{c^2})$

1. 纵波: $\vec{P} \parallel \vec{q}$

$\vec{E}_L = -\frac{1}{\epsilon_0} \vec{P}$ 使离子所受恢复力变大, $\omega_L \uparrow$

2. 横波: $\vec{P} \perp \vec{q}$

$$\vec{E}_T = \frac{\omega^2}{\epsilon_0(c^2 q^2 - \omega^2)} \vec{P}$$

$$\left\{ \begin{array}{l} \omega > cq: \omega_T \uparrow \\ \omega = cq: \text{共振} \\ \omega < cq: \omega_T \downarrow \end{array} \right.$$

二、长光学波的宏观运动方程

1. 运动方程.

设每个元胞中有 m_+ , m_- 两个离子, 相对位移为 $\vec{u}_+ - \vec{u}_-$

令 $\vec{w} = \sqrt{\frac{m}{2}} (\vec{u}_+ - \vec{u}_-)$

动能密度: $T = \frac{1}{2} \dot{\vec{w}}^2$

势能密度: $V = V_{弹性} + V_{极化}$

$$= - \int \vec{F} \cdot d\vec{w} - \int \vec{P} \cdot d\vec{E}$$

写出能量.

$$\vec{E} = \epsilon_0 \vec{W}$$

$$\vec{P}_{\text{eff}} = \epsilon_{12} \vec{W}$$

$$\vec{P}_{\text{ex}} = \epsilon_{22} \vec{E}$$

$$= - \int \vec{F} \cdot d\vec{W} - \int (\vec{P}_{\text{eff}} + \vec{P}_{\text{ex}}) \cdot d\vec{E}$$

$$\approx - \int \epsilon_{11} \vec{W} \cdot d\vec{W} - \int (\epsilon_{12} \vec{W} + \epsilon_{22} \vec{E}) \cdot \vec{E}$$

$$= - \frac{1}{2} \epsilon_{11} \vec{W}^2 - \epsilon_{12} \vec{W} \cdot \vec{E} - \frac{1}{2} \epsilon_{22} \vec{E}^2$$

$$L = T - U = \frac{1}{2} \dot{\vec{W}}^2 + \frac{1}{2} \epsilon_{11} \vec{W}^2 + \epsilon_{12} \vec{W} \cdot \vec{E} + \frac{1}{2} \epsilon_{22} \vec{E}^2$$

共轭动量: $\vec{P} = \frac{\partial L}{\partial \dot{\vec{W}}} = \dot{\vec{W}}$

$$H = L + U = \frac{1}{2} \dot{\vec{W}}^2 - \frac{1}{2} \epsilon_{11} \vec{W}^2 - \epsilon_{12} \vec{W} \cdot \vec{E} - \frac{1}{2} \epsilon_{22} \vec{E}^2$$

$\Rightarrow$ Hamilton 方程: $\dot{q} = \vec{W}$, $\vec{P} = \frac{\partial L}{\partial \dot{\vec{W}}} = \dot{\vec{W}}$, $\dot{\vec{P}} = - \frac{\partial H}{\partial \vec{W}}$

$$\begin{cases} \dot{\vec{W}} = - \frac{\partial H}{\partial \vec{P}} = \epsilon_{11} \vec{W} + \epsilon_{12} \vec{E} \\ \dot{\vec{P}} = \epsilon_{12} \vec{W} + \epsilon_{22} \vec{E} \end{cases} \quad \begin{bmatrix} \dot{\vec{W}} \\ \dot{\vec{P}} \end{bmatrix} = \vec{M} \begin{bmatrix} \vec{W} \\ \vec{E} \end{bmatrix}$$

2. $\epsilon_{11}, \epsilon_{12}, \epsilon_{22}$ 均取定.

① $\vec{E}$ 为恒定静电场 ($\dot{\vec{W}} = 0$)

$$\epsilon_{11} \vec{W} = - \epsilon_{12} \vec{E}$$

$$\vec{P} = (\epsilon_{22} - \frac{\epsilon_{12}^2}{\epsilon_{11}}) \vec{E} = [\epsilon(\omega) - 1] \vec{E}$$

② $\vec{E}$ 为高频电场 ($\dot{\vec{W}} \neq 0$)

$$\vec{P} = \epsilon_{22} \vec{E} = [\epsilon(\omega) - 1] \vec{E}$$

3. 共振频率 ω_{TO} , ω_{LO}

在长波情况下, ω 与 q 无关, 因此有共振频率 ω_{LO} , ω_{TO}

对声子而言, 由于纵向振动时静电场增加了恢复力, $\omega_{LO} > \omega_{TO}$

Maxwell 方程

$$\vec{W} = \epsilon_{11} \vec{W} + \epsilon_{12} \vec{E}$$

$$\vec{P} = \epsilon_{12} \vec{W} + \epsilon_{22} \vec{E}$$

$$\vec{\nabla} \cdot \vec{D} = \vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) = 0$$

$$\vec{\nabla} \times \vec{E} = 0$$

位移 $\rightarrow$ 横场 + 纵场

$$W = W_T + W_L$$

$$\nabla \times W_T \neq 0, \quad \nabla \cdot W_T = 0 \quad \text{可以用 } \nabla \leftrightarrow \vec{k} \text{ 来记}$$

$$\nabla \times W_L = 0, \quad \nabla \cdot W_L \neq 0$$

$$\nabla \times (\ddot{W}_T + \ddot{W}_L) = b_{11} \nabla \times (W_T + W_L) + b_{12} \nabla \times E$$

$$\nabla \times \ddot{W}_T = b_{11} \nabla \times W_T$$

$$\text{解得 } W_{T0}^2 = -b_{11} = \omega_0^2 \quad (b_{11} = \frac{F}{W})$$

横场有驻频率只与弹性恢复力有关。

$$\nabla \cdot (\ddot{W}_T + \ddot{W}_L) = b_{11} \nabla \cdot (W_T + W_L) + b_{12} \nabla \cdot E$$

$$\nabla \cdot P = b_{21} \nabla \cdot (W_T + W_L) + b_{22} \nabla \cdot E$$

$$\nabla \cdot \ddot{W}_L = b_{11} \nabla \cdot W_L + b_{12} \nabla \cdot E$$

$$-\epsilon_0 \nabla \cdot E = b_{21} \nabla \cdot W_L + b_{22} \nabla \cdot E \rightarrow \nabla \cdot E = \frac{-b_{21}}{b_{22} + \epsilon_0} \nabla \cdot W_L$$

$$\nabla \cdot \ddot{W}_L = (b_{11} - \frac{b_{12}^2}{b_{22} + \epsilon_0}) \nabla \cdot W_L$$

$$\omega_{L0}^2 = \frac{b_{12}^2}{b_{22} + \epsilon_0} - b_{11} = \frac{b_{12}^2 - b_{11}(b_{22} + \epsilon_0)}{\epsilon(\infty) - 1 + 1}$$

$$= \frac{1}{\epsilon(\infty)} \left\{ b_{12}^2 - b_{11} \left[\epsilon(\infty) - 1 + 1 + \frac{b_{11}}{b_{11}} \right] \right\}$$

$$= \frac{\epsilon(0)}{\epsilon(\infty)} (-b_{11}) = \frac{\epsilon(0)}{\epsilon(\infty)} \omega_{T0}^2$$

$$LST \text{ 关系: } \frac{\omega_{L0}^2}{\omega_{T0}^2} = \frac{\epsilon(0)}{\epsilon(\infty)}$$

$\epsilon(0)$: 所有带电粒子的响应。

$$\epsilon(0) > \epsilon(\infty), \quad \omega_{L0} > \omega_{T0}$$

$\epsilon(\infty)$: 电子的响应 (极化)

四、极化激元 (耦合声子 - 光子场的量子)

在布里渊区中心附近小波矢的情况下, 声子 / 光子频率、波矢近似相等

Huang & Maxwell 方程: $\mu = 1, \vec{j} = 0, \rho = 0$

$$\ddot{W} = b_{11} W + b_{12} E$$

$$\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

$$\nabla \times \vec{H} = \frac{\partial}{\partial t} (\epsilon_0 \vec{E} + \vec{P})$$

$$\nabla \cdot \vec{P} = \nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = 0$$

$$\nabla \cdot \vec{H} = 0$$

代入波动解: $\vec{W} = \vec{W}_0 e^{i(\vec{q} \cdot \vec{r} - \omega t)}$

$$\vec{P} = \vec{P}_0 e^{i(\vec{q} \cdot \vec{r} - \omega t)}$$

$$\vec{E} = \vec{E}_0 e^{i(\vec{q} \cdot \vec{r} - \omega t)}$$

$$\vec{H} = \vec{H}_0 e^{i(\vec{q} \cdot \vec{r} - \omega t)}$$

$$-\omega^2 \vec{W}_0 = b_{11} \vec{W}_0 + b_{12} \vec{E}_0 \quad (1)$$

$$\vec{P}_0 = b_{21} \vec{W}_0 + b_{22} \vec{E}_0 \quad (2)$$

$$i\vec{q} \times \vec{E}_0 = i\omega \mu_0 \vec{H}_0 \quad (3)$$

$$i\vec{q} \times \vec{H}_0 = -i\omega (\epsilon_0 \vec{E}_0 + \vec{P}_0) \quad (4)$$

$$i\vec{q} \cdot \vec{P}_0 = i\vec{q} \cdot (\epsilon_0 \vec{E}_0 + \vec{P}_0) = 0 \quad (5)$$

$$i\vec{q} \cdot \vec{H}_0 = 0 \quad (6)$$

$$(1), (2) \rightarrow \vec{P}_0 = (b_{22} - \frac{b_{12}^2}{b_{11} + \omega^2}) \vec{E}_0$$

$$(3) \rightarrow (\vec{q} \cdot \vec{E}_0) (\epsilon_0 + b_{22} - \frac{b_{12}^2}{b_{11} + \omega^2}) = 0$$

当此处使 $\vec{q} \cdot \vec{E}_0 \neq 0$

$$\epsilon_0 + b_{22} - \frac{b_{12}^2}{b_{11} + \omega^2} = 0$$

$$\omega^2 = \omega_{LO}^2 = \frac{b_{12}^2}{\epsilon_0 + b_{22}} - b_{11} = \frac{\epsilon_{(\infty)}}{\epsilon_{(0)}} \omega_{PO}^2$$

∴ 此声子不与电磁场耦合

当对横波 $\vec{q} \cdot \vec{E}_0 = 0$, $\vec{q} \perp \vec{E}_0 \perp \vec{H}_0$

$$(3) \rightarrow q E_0 = \mu_0 \omega H_0$$

$$(4) \rightarrow \omega (\epsilon_0 + b_{22} - \frac{b_{12}^2}{b_{11} + \omega^2}) E_0 = q H_0$$

(E_0, H_0) 有非零解

$$\left| \begin{matrix} q & -\mu_0 \omega \\ \omega(\epsilon_0 + b_{11} - \frac{b_{12}^2}{b_{11} + \omega^2}) & -q \end{matrix} \right|$$

$$= \omega^2 \mu_0 (\epsilon_0 + b_{11} - \frac{b_{12}^2}{b_{11} + \omega^2}) - q^2 = 0$$

$$\Rightarrow \frac{c^2 q^2}{\omega^2} = \epsilon(\omega) \frac{\omega_{L0}^2 - \omega^2}{\omega_{T0}^2 - \omega^2}$$

解得极化波元的色散关系

$$\omega_{\pm}^2(q) = \frac{1}{2} \left\{ \omega_{L0}^2 + \frac{c^2 q^2}{\epsilon(\infty)} \pm \left[\omega_{L0}^4 + \frac{c^4 q^4}{\epsilon^2(\infty)} + \frac{2c^2 q^2}{\epsilon(\infty)} (\omega_{L0}^2 - \omega_{T0}^2) \right]^{1/2} \right\}$$

1) $q \rightarrow 0$

$\omega_+ \approx \omega_{L0}$ 类声子

$\omega_- \approx \frac{c}{\sqrt{\epsilon(\infty)}} q$ 类光子

2) $q \rightarrow \infty$

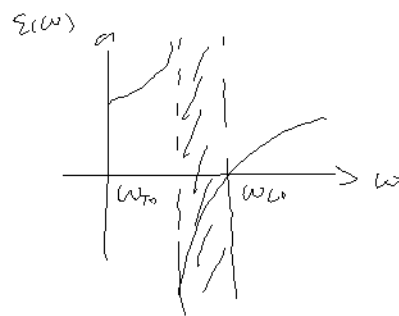
$\omega_+ \approx \frac{c}{\sqrt{\epsilon(\infty)}} q$ 类光子

$\omega_- \approx \omega_{T0}$ 类声子

3) $\omega \in (\omega_{T0}, \omega_{L0})$ 是频率禁区

$$\epsilon(\omega) = \frac{c^2 q^2}{\omega^2} = \epsilon(\infty) \frac{\omega_{L0}^2 - \omega^2}{\omega_{T0}^2 - \omega^2}$$

此时 q 为虚数，波将衰减。



§ 3.6 非完整晶格的振动 局域模

一、一维单原子链的扩展模式

色散关系 $u_l = A e^{i[q_1 a - \omega(q) t]}$, $q \in \text{BZ}$

$$\omega(q) = \omega_m |\sin(\frac{1}{2} q a)|, \quad \omega_m = 2 \sqrt{\frac{f}{m}}$$

(1) $\omega \leq \omega_m$, 扩展模式

(2) $\omega > \omega_m$, 局域模式

$$\frac{\omega}{\omega_m} = \sin(\frac{1}{2} q a) = \sin\left[\frac{1}{2} (q_1 + i q_2) a\right] \quad q = q_1 + i q_2, \quad q_2 \neq 0$$

$$= \sin(\frac{1}{2} q_1 a) \cosh(\frac{1}{2} q_2 a) + i \cos(\frac{1}{2} q_1 a) \sinh(\frac{1}{2} q_2 a)$$

$$\omega, \omega_m \text{ 为实数} \rightarrow \cosh(\frac{1}{2} q_2 a) = 0$$

$$\frac{1}{2} q_1 a = (h \pm \frac{1}{2}) \pi,$$

$$q_1 = \frac{2\pi}{a} (h \pm \frac{1}{2}) = K_h \pm \frac{\pi}{a} = \frac{\pi}{a} \quad (q_1 \in \text{BZ})$$

$$\rightarrow q_2 = \frac{2}{a} \operatorname{arccosh}\left(\frac{\omega}{\omega_m}\right)$$

$$q = \frac{\pi}{a} + i \frac{2}{a} \operatorname{arccosh}\left(\frac{\omega}{\omega_m}\right)$$

$$u_l = A e^{i[(q_1 + i q_2) l a - \omega(q) t]}$$

$$= A e^{-2 \operatorname{arccosh}(\frac{\omega}{\omega_m}) l} e^{i K l} e^{-i \omega(q) t}$$

二、含单个缺陷的一维单原子链

$$m_l = \begin{cases} m & l \neq 0 \\ m' & l = 0 \end{cases} \quad \epsilon = \frac{m' - m}{m}$$

$$H = T + V$$

$$= \frac{1}{2} m \sum_i \dot{u}_i^2 + \frac{1}{2} (m' - m) \dot{u}_0^2 + \frac{1}{2} \beta \sum_i (u_{i+1} - u_i)^2$$

在扩展模的基下展开:

$$u_i = \frac{1}{\sqrt{N}} \sum_q \alpha_q e^{iqia}$$

$$\text{其中 } \alpha_q = \frac{1}{\sqrt{N}} \sum_i u_i e^{-iqia}$$

$$\begin{aligned} H &= \frac{1}{2} m \cdot \frac{1}{N} \sum_q \sum_{q'} \alpha_q^\dagger \alpha_{q'} e^{i(q+q')ia} + \frac{1}{2} (m' - m) \frac{1}{N} \sum_{q,q'} \alpha_q^\dagger \alpha_{q'} \\ &\quad + \frac{1}{2} \beta \frac{1}{N} \sum_i \left\{ \sum_q \alpha_q \left| e^{iq(1+i)a} - e^{iqia} \right| \right\} \left\{ \sum_{q'} \alpha_{q'} \left| e^{iq'(1+i)a} - e^{iq'ia} \right| \right\} \\ &= \frac{m}{2} \sum_{q,q'} \left(\delta_{q,-q'} + \frac{\epsilon}{N} \right) \alpha_q^\dagger \alpha_{q'} \\ &\quad + \frac{\beta}{2N} \sum_q \sum_{q'} \alpha_q \alpha_{q'} e^{i(q+q')ia} (e^{iqa} - 1)(e^{iq'a} - 1) \\ &= \frac{m}{2} \sum_{q,q'} \left(\delta_{q,-q'} + \frac{\epsilon}{N} \right) \alpha_q^\dagger \alpha_{q'} + \left[\frac{\beta}{m} (e^{iqa} - 1)(e^{-iq'a} - 1) \right] \delta_{q,-q'} \alpha_q \alpha_{q'} \\ &= \frac{m}{2} \left[\sum_q \alpha_q^\dagger \alpha_{-q} + \frac{\epsilon}{N} \left(\sum_q \alpha_q^\dagger \right)^2 \right] + \frac{m}{2} \cdot \sum_q \frac{2\beta}{m} [1 - \cos(qa)] \alpha_q \alpha_{-q} \\ L &= \frac{m}{2} \left[\sum_q \alpha_q^\dagger \alpha_{-q} + \frac{\epsilon}{N} \left(\sum_q \alpha_q^\dagger \right)^2 \right] - \frac{m}{2} \cdot \sum_q \frac{2\beta}{m} [1 - \cos(qa)] \alpha_q \alpha_{-q} \end{aligned}$$

$$p = \frac{\partial L}{\partial \dot{\alpha}_q} = m (\dot{\alpha}_{-q} + \frac{\epsilon}{N} \sum_{q'} \dot{\alpha}_{q'})$$

$$\dot{p} = - \frac{\partial H}{\partial \alpha_q} = -m \frac{2\beta}{m} [1 - \cos(qa)] \alpha_{-q}$$

$$\Rightarrow \ddot{\alpha}_{-q} + \frac{\epsilon}{N} \sum_{q'} \ddot{\alpha}_{q'} + \frac{2\beta}{m} [1 - \cos(qa)] \alpha_{-q} = 0$$

代入波动解

$$\alpha_q = C_q e^{i\omega t}$$

$$-\omega^2 \left(C_{-q} + \frac{\epsilon}{N} \sum_{q'} C_{q'} \right) + \frac{2\beta}{m} [1 - \cos(qa)] C_{-q} = 0$$

$$\text{解得 } C_{-q} = \frac{\sum_{q'} \omega^2}{N \left\{ \frac{2\beta}{m} [1 - \cos(qa)] - \omega^2 \right\}} \sum_{q'} C_{q'}$$

$$\Rightarrow \sum_q C_q = S$$

$$= \sum_q \frac{\sum_{q'} \omega^2}{N \left\{ \frac{2\beta}{m} [1 - \cos(qa)] - \omega^2 \right\}} S$$

$$\Rightarrow \frac{2}{9} \mathcal{N} \left\{ \frac{2\beta}{\omega_m} \left[1 - \cos(qa) \right] - \omega^2 \right\} = 1$$

杂质的引入 / 无界将导致局域化

三. 局域模

在 $l=0$ 上有一轻杂质原子

$$\begin{cases} m' \ddot{u}_0 - \beta(u_1 + u_{-1} - 2u_0) = 0 \\ m \ddot{u}_l - \beta(u_{l+1} + u_{l-1} - 2u_l) = 0 \quad (l \neq 0) \end{cases}$$

设 $u_l \sim e^{-i\omega t}$

$$\frac{\beta}{m} = \frac{1}{4} \omega_m^2 \frac{m}{m'}$$

$$= \frac{\omega_m^2}{4} \frac{1}{1-\varepsilon}$$

$$\begin{cases} -m' \omega^2 u_0 - \beta(u_1 + u_{-1} - 2u_0) = 0 \\ -m \omega^2 u_l - \beta(u_{l+1} + u_{l-1} - 2u_l) = 0 \end{cases}$$

$$\begin{cases} \left[\frac{4\omega^2(1-\varepsilon)}{\omega_m^2} - 2 \right] u_0 + u_1 + u_{-1} = 0 & ① \\ \left(\frac{4\omega^2}{\omega_m^2} - 2 \right) u_l + u_{l+1} + u_{l-1} = 0 \quad (l \neq 0) & ② \end{cases}$$

其中 $\varepsilon = \frac{m-m'}{m}$, $\omega_m = 2\sqrt{\frac{\beta}{m}}$,

对称性: $u \rightarrow$ 对称解 $u_l = u_{-l}$ / 反对称解 $u_l = -u_{-l}$ ($u_0 = 0$)

对称解: 设 $u_l = A \lambda^{|l|}$, —— 舍去

$$A \left[\left(\frac{4\omega^2}{\omega_m^2} - 2 \right) \lambda^{|l|} + \lambda^{|l+1|} + \lambda^{|l-1|} \right] = 0$$

$$\lambda^2 + \left(\frac{4\omega^2}{\omega_m^2} - 2 \right) \lambda + 1 = 0$$

$$\Delta = 4 \left[2 \left(\frac{\omega}{\omega_m} \right)^2 - 1 \right]^2 - 4 = 16 \left(\frac{\omega}{\omega_m} \right) \left(\frac{\omega}{\omega_m} - 1 \right)$$

① $\omega < \omega_m$, $\Delta < 0$, 共轭复根.

$$\lambda_{\pm} = e^{\pm iqa}$$

$$u_l = A \lambda_+^{|l|} + B \lambda_-^{|l|} = A e^{iqa|l|} + B e^{-iqa|l|}$$

$$\omega^2(q) = \omega_m^2 \sin^2(\frac{1}{2}qa)$$

$$\text{代入 ①} \quad \frac{4\omega^2(1-\varepsilon)}{\omega_m^2} (A+B) + 2(Ae^{iqa} + Be^{-iqa}) = 0$$

$$u_1 \in \mathbb{R} \rightarrow A = B^*$$

$$\text{令 } A = \frac{C}{2} e^{-i\delta}, B = \frac{C}{2} e^{i\delta}$$

$$\Rightarrow u_1 = C \cos(qa(1-\delta))$$

$$\text{其中 } \delta = \arctan\left[\varepsilon \tan\left(\frac{1}{2}qa\right)\right]$$

② $\omega > \omega_m$, $\varepsilon > 0$, 有实根解.

$$u_1 = A \lambda^{(1)}$$

$$\begin{cases} 2\lambda^2 + \frac{4\omega^2}{\omega_m^2} (1-\varepsilon) - 2 = 0 \\ \lambda^2 + (\frac{4\omega^2}{\omega_m^2} - 2)\lambda + 1 = 0 \end{cases}$$

$$\text{解得 } \lambda = 1 \text{ (舍去)} \quad \lambda = \frac{1-\varepsilon}{1+\varepsilon}$$

$$\begin{aligned} u_1 &= A (-1)^{(1)} \left(\frac{1-\varepsilon}{1+\varepsilon}\right)^{(1)} e^{-i\omega z} \\ &= (-1)^{(1)} A e^{-\alpha(1)} e^{-i\omega z} \end{aligned}$$

$$\text{其中 } \alpha = m \left\langle \frac{1+\varepsilon}{1-\varepsilon} \right\rangle$$

§ 3.7 晶格比热容

晶格比热容 = 晶格比热容 + 电子比热容.

一、声子态密度.

$$U(V, T) = \sum_{\vec{q}, s} \left[n_{\vec{q}, s}(T) + \frac{1}{2} \right] \hbar \omega_s(\vec{q})$$

$$= \sum_{\vec{q}, s} \left[e^{\hbar \omega_s(\vec{q}) / kT} - 1 + \frac{1}{2} \right] \hbar \omega_s(\vec{q})$$

$$C_V(T) = \frac{\partial U(V, T)}{\partial T}$$

$$= \sum_{\vec{q}, s} \frac{\partial}{\partial T} \left[e^{\hbar \omega_s(\vec{q}) / kT} - 1 + \frac{1}{2} \right] \hbar \omega_s(\vec{q})$$

$$= \sum_{\vec{q}, s} k \cdot \frac{[\hbar \omega_s(\vec{q})]^2}{(e^{\hbar \omega_s(\vec{q}) / kT} - 1)^2} e^{\hbar \omega_s(\vec{q}) / kT}$$

$$= \sum_{\vec{q}, s} C[\omega_s(\vec{q})]$$

$N \rightarrow \infty$ 时, $\vec{q}$ 准连续分布.

$$= \sum_s \int_{\Omega^*} \frac{V}{(2\pi)^3} C[\omega_s(\vec{q})] d^3\vec{q}$$

$$= \int_0^\infty \rho(\omega) C(\omega) d\omega$$

$$\text{其中 } \rho(\omega) = \frac{V}{(2\pi)^3} \sum_s \int_{\Omega^*} d^3\vec{q} \delta[\omega - \omega_s(\vec{q})]$$

$$\begin{aligned}
 \int_0^\infty \rho(\omega) d\omega &= \int_0^\infty \frac{V}{(2\pi)^3} \left\{ \frac{2}{3} \int_{\Omega^*} d^3 \vec{q} \delta[\omega - \omega_s(\vec{q})] \right\} d\omega \\
 &= \frac{V}{(2\pi)^3} \frac{2}{3} \int_{\Omega^*} d^3 \vec{q} \int_0^\infty d\omega \delta[\omega - \omega_s(\vec{q})] \\
 &= \frac{V}{(2\pi)^3} \frac{2}{3} \int_{\Omega^*} d^3 \vec{q} \cdot 1 \\
 &= \frac{V}{(2\pi)^3} \cdot \Omega^* \cdot \frac{2}{3} \quad V = N\Omega, \quad \Omega = \frac{(2\pi)^3}{\Omega^*} \\
 &= \frac{2}{3} N
 \end{aligned}$$

$$\rho(\omega) = \frac{V}{(2\pi)^3} \frac{2}{3} \int_{\Omega^*} d^3 \vec{q} \delta[\omega - \omega_s(\vec{q})]$$

① 在 $[\frac{\omega}{2}, \frac{3\omega}{2}]$ 内

将 $\vec{q} \in \Omega^* \cap [\frac{\omega}{2}, \frac{3\omega}{2}]$ 为等频率面 S_ω

$$d^3 \vec{q} = dS_\omega d\vec{q}_\perp, \quad d\omega = |\nabla_{\vec{q}} \omega_s(\vec{q})| \cdot d\vec{q}_\perp$$

$$\begin{aligned}
 \rho(\omega) &= \frac{V}{(2\pi)^3} \frac{2}{3} \int dS_\omega d\vec{q}_\perp \delta[\omega - \omega_s(\vec{q})] \\
 &= \frac{V}{(2\pi)^3} \frac{2}{3} \int dS_\omega d\omega_s \frac{\delta[\omega - \omega_s(\vec{q})]}{|\nabla_{\vec{q}} \omega_s(\vec{q})|} \\
 &= \frac{V}{(2\pi)^3} \frac{2}{3} \int \frac{dS_\omega}{|\nabla_{\vec{q}} \omega_s(\vec{q})|}
 \end{aligned}$$

② 在 $[\frac{\omega}{2}, \frac{3\omega}{2}]$ 内

$$\rho_{2D}(\omega) = \frac{S}{(2\pi)^2} \frac{2}{3} \int \frac{d\omega}{|\nabla_{\vec{q}} \omega_s(\vec{q})|}$$

③ 在 $[\frac{\omega}{2}, \frac{3\omega}{2}]$ 内

$$\rho_{1D}(\omega) = \frac{L}{2\pi} \frac{2}{3} \frac{2}{|\partial\omega/\partial q|}$$

三. Einstein 模型, 比热容

设晶格中所有原子以 $\omega = \omega_E$ 振动 (对应光学支)

$$\rho(\omega) = \frac{V}{(2\pi)^3} \sum_s \int_{\Omega^*} d^3q \delta(\omega - \omega_E)$$

$$= \frac{V}{(2\pi)^3} \Omega^* \cdot \delta(\omega - \omega_E) = 3nN \delta(\omega - \omega_E)$$

$$C_V^E(T) = \int_0^\infty \rho(\omega) C(\omega) d\omega = 3nN C(\omega_E)$$

$$= 3nN \frac{\left(\frac{\hbar\omega_E}{kT}\right)^2}{\left(e^{\frac{\hbar\omega_E}{kT}} - 1\right)^2} k e^{\frac{\hbar\omega_E}{kT}}$$

$$= 3nN \frac{(T_E/T)^2}{(e^{T_E/T} - 1)^2} k e^{T_E/T}$$

$$\sim \begin{cases} 3nNk & (T \gg T_E) \\ 3nNk e^{-\frac{T_E}{T}} & (T \ll T_E) \end{cases}$$

四. Debye 模型, 比热容.

将晶体作为连续介质处理

$$\omega_s(\vec{q}) = c_s |\vec{q}|$$

Debye 截止频率, 布里渊区 $\sim$ Debye 球:

$$q \leq q_0, \quad \frac{4}{3}\pi q_0^3 = \Omega^*, \quad q_0 = \left(\frac{3\Omega^*}{4\pi}\right)^{1/3} = \left(\frac{6N\pi^2}{V}\right)^{1/3}$$

$$\omega \leq \omega_D = 0.90, \quad \bar{c}^2 = \frac{1}{c_1^2} + \frac{1}{c_2^2} \quad (\text{两模一致})$$

声子谱密度.

$$\rho_D(\omega) = \frac{V}{(2\pi)^3} \sum_s \int \frac{dS_{\vec{\omega}}}{|\vec{v}_s(\vec{q})|} \quad S = 1, 2, 3 \quad (\vec{z} \text{ 支})$$

$$= \frac{V}{(2\pi)^3} \sum_s \frac{1}{c_s} \int dS_{\vec{\omega}} = \frac{V}{(2\pi)^3} \sum_s \frac{4\pi \omega_s^2}{c_s}$$

$$= \frac{V}{2\pi^2} \sum_s \frac{\omega_s^2}{c_s^3} = \frac{3V\omega^2}{2\pi^2 c}$$

$$= \frac{3V\omega^2}{2\pi^2 \omega_D^3} \cdot \rho_D^3 = \frac{9N\omega^2}{\omega_D^3} \quad (\omega \in [0, \omega_D])$$

$$= 0 \quad \omega > \omega_D$$

比热容: $C_V = \int_0^{\omega_D} \rho_D(\omega) C(\omega) d\omega$

$$= \int_0^{\omega_D} \frac{9N\omega^2}{\omega_D^3} \cdot k \cdot \frac{\left(\frac{\hbar\omega}{kT}\right)^2 e^{\hbar\omega/kT}}{(e^{\hbar\omega/kT} - 1)^2} d\omega$$

Debye 温度: $\theta_D = \frac{\hbar\omega_D}{k}$

$$= \frac{9N}{\omega_D^3} \left(\frac{kT}{\hbar}\right)^3 \int_0^{\theta_D/T} \frac{z^4 e^z}{(e^z - 1)^2} dz$$

$$= 9Nk \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D/T} \frac{z^4 e^z}{(e^z - 1)^2} dz$$

$$= 3Nk f\left(\frac{T}{\theta_D}\right)$$

§ 3.8 晶格状态方程和热膨胀

一、自由能和格林艾森方程

1.

内能 = 基态 + 热振动的能量.

$$U(T, V) = U^0(V) + U^v(V, T)$$

$$\text{其中 } U^v(V, T) = \sum_{\vec{q}, s} n_{\vec{q}, s}(T) \hbar \omega_s(\vec{q})$$

$$= \int_0^\infty \frac{\hbar \omega}{e^{\frac{\hbar \omega}{kT}} - 1} \rho(\omega) d\omega$$

2.

$$\begin{aligned} \text{自由能 } F(T, V) &= U - TS = U^0(V) + U^v(V, T) - TS \\ &= F^0(V) + F^v(T, V) \end{aligned}$$

$$\text{其中 } F^v(V, T) = -kT \ln Z^v$$

系统由 $3N$ 个 $\varepsilon_{\vec{q}, s} = n_{\vec{q}, s} \hbar \omega_s(\vec{q})$ 的谐振子组成.

$$Z^v = \sum_{n_{\vec{q}, s}} e^{-\frac{\varepsilon_{\vec{q}, s}}{kT}} = \prod_{\vec{q}, s} \frac{1}{1 - e^{-\frac{\hbar \omega_s(\vec{q})}{kT}}}$$

$$Z^v = \prod_{\vec{q}, s} Z_{\vec{q}, s}^v = \prod_{\vec{q}, s} \frac{1}{1 - e^{-\frac{\hbar \omega_s(\vec{q})}{kT}}}$$

$$\hookrightarrow F^v(V, T) = kT \sum_{\vec{q}, s} \ln(1 - e^{-\frac{\hbar \omega_s(\vec{q})}{kT}})$$

$$= kT \int_0^\infty \ln(1 - e^{-\frac{\hbar \omega}{kT}}) \rho(\omega) d\omega$$

3.

系统状态方程：

$$\begin{aligned} P &= - \left[\frac{\partial F(T, V)}{\partial V} \right]_T \\ &= - \frac{dU^0(V)}{dV} - \frac{\partial F^V(T, V)}{\partial V} \\ &= - \frac{dU^0(V)}{dV} - kT \sum_{\vec{q}s} \frac{e^{-\frac{\hbar \omega_s(\vec{q})}{kT}}}{1 - e^{-\frac{\hbar \omega_s(\vec{q})}{kT}}} \left(- \frac{\hbar}{kT} \right) \frac{d\omega_s(\vec{q})}{dV} \\ &= - \frac{dU^0(V)}{dV} - \hbar \sum_{\vec{q}s} \frac{e^{-\frac{\hbar \omega_s(\vec{q})}{kT}}}{1 - e^{-\frac{\hbar \omega_s(\vec{q})}{kT}}} \frac{\omega_s(\vec{q})}{V} \frac{d(\ln \omega_s(\vec{q}))}{d \ln V} \end{aligned}$$

近似认为 $-\frac{d \ln \omega_s(\vec{q})}{d \ln V} \approx \sigma$ ，格林艾森常数。

$$\begin{aligned} P &= - \frac{dU^0(V)}{dV} + \frac{\sigma}{V} \sum_{\vec{q}s} \hbar \omega_s(\vec{q}) \\ &= - \frac{dU^0(V)}{dV} + \sigma \frac{U^V(T, V)}{V} = P_{th} + P_{ph} \end{aligned}$$

估算， $\sigma \approx 1$ ， $U^V(T, V) \approx 3 N_A kT = 3RT = 3 \cdot P_g V_g$ 。

$$\begin{aligned} P_{ph} &= 3\sigma P_g \frac{V_g}{V_s} \quad P_g \sim 10^5 \text{ Pa}, \quad \frac{V_g}{V_s} \sim 10^2 \\ &\sim 10^8 \text{ Pa} \end{aligned}$$

二. 热膨胀

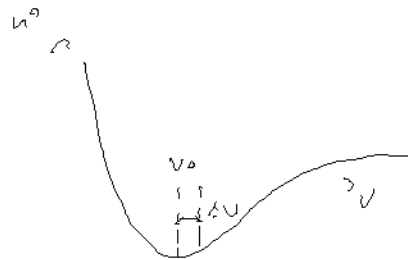
1. 定性解释。

$$P \approx 0$$

$$\frac{dU^0(V)}{dV} = \sigma \frac{U^V(T, V)}{V} \quad \left. \begin{array}{l} (T=0) \\ (T>0) \end{array} \right\} \begin{array}{l} = 0 \\ > 0 \end{array}$$

$$T=0, \quad \frac{dU^0(V)}{dV} = 0, \quad V=V_0$$

$$T>0, \quad \frac{dU^0(V)}{dV} > 0 \Rightarrow V = V_0 + \Delta V > V_0$$



2. 热膨胀的格林艾森关系。

$$(P + \frac{dU^0(V)}{dV}) V = \sigma U^V(T, V)$$

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{\partial}{\partial T} \frac{\partial U(T, V)}{\partial V} = \sigma C_V$$

$$\left(\frac{\partial P}{\partial T}\right)_S \left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial V}{\partial P}\right)_T = -1$$

$$- \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial P}{\partial V}\right)_T = \sigma C_V$$

$$\sigma = \frac{B \alpha V}{C_V} \sim \{1, 3\}$$

压缩性模量 $B = -V \left(\frac{\partial P}{\partial V}\right)_T$

热膨胀系数 $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P$

三. 热膨胀与非谐效应

以一维单原子链为例

$$\omega^2(q) = \frac{4\beta}{m} \sin^2\left(\frac{1}{2}qa\right)$$

$$\sigma = - \frac{d \ln \omega}{d \ln L} = - \frac{2}{N} \frac{d}{da} \left[\frac{4\beta}{m} \sin^2\left(\frac{1}{2}qa\right) \right]$$

$$= - \frac{2}{N} \cdot \frac{4\beta}{m} 2 \sin\left(\frac{1}{2}qa\right) \cos\left(\frac{1}{2}qa\right) \left(\frac{1}{2}q\right)$$

$$= - \frac{4\beta q}{Nm} \sin(qa)$$